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POLICY ADAPTATION BASED ON IMPORTANCE SAMPLING

Abstract

A system and a method are disclosed for scheduling, the method includes determining a second state distribution based on executing a first scheduling policy in a second environment, generating a second scheduling policy based on a first policy gradient associated with a first state distribution and the second state distribution, the first state distribution being associated with the first scheduling policy and a first environment, different from the second environment, and controlling a scheduling process based on the second scheduling policy.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the priority benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 63/560,287, filed on Mar. 1, 2024, the disclosure of which is incorporated by reference in its entirety as if fully set forth herein.

TECHNICAL FIELD

[0002] The disclosure generally relates to manufacturing systems and methods. More particularly, the subject matter disclosed herein relates to improvements to scheduling by performing policy adaptation based on importance sampling.

SUMMARY

[0003] Manufacturing facilities may face the challenge of efficiently producing a diverse range of products. For example, a number of product lots (e.g., product types) may be in a range of hundreds of thousands. Each product lot may undergo numerous processes across multiple machines. Efficient coordination of the machines, products, resources, and/or the like of a manufacturing system is important for achieving higher productivity and utilization. Experts with extensive domain expertise and years of training may be able to produce reasonable manufacturing scheduling policies. However, adapting existing scheduling policies to changing scenarios and changing evaluation criteria (e.g., evaluation metrics) based on human experience and expertise alone may be inadequate for achieving satisfactory results. As used herein, a “scenario” refers to a configuration (e.g., a status) of an environment (e.g., a manufacturing environment), which may be represented by a variety of parameters associated with the environment and collected in a dataset. Some examples of scenarios include states of products, resources, and/or machines and logistics associated with the products, resources, and/or machines.

[0004] Reinforcement learning (RL) may be used to help determine improved scheduling. For example, a scheduling policy may be trained in a training environment on a simulator (e.g., a digital simulator) using a historic dataset to develop (e.g., to train) the scheduling policy for use in controlling a scheduling process. The historic dataset may include gaps in scenario coverage when the trained scheduling policy is deployed. For example, the historic dataset may have been collected over an incomplete time period (e.g., only part of a year), or various parameters may have changed since the data for the historic dataset was collected.

[0005] Misalignment between training data and deployment scenarios may reduce (e.g., may significantly reduce) system performance. For example, trainings may not be effective for achieving satisfactory performance when a distribution in a state visitation (e.g., a steady-state distribution) in a training environment is not similar to (e.g., not sufficiently similar to) a distribution in the state visitation (e.g., a steady-state distribution) in a deployment environment. As used herein, a “state distribution” refers to a probability that a given state will be visited or will occur in reinforcement learning.

[0006] It may be suitable for systems to be able to perform policy adaptation using computers with limited computational power. For example, industrial computers used in deployment may include general purpose processors (e.g., central processing units (CPUs)) that are less capable than specialized processors (e.g., graphics processing units (GPUs)) at performing training (e.g., complex training). Strict deployment time limits (e.g., short deployment time limits) may further restrict training and adaptation. For example, a scheduling policy may need to be deployed and updated within a matter of minutes (e.g., several minutes).

[0007] In some embodiments of the present disclosure, policy adaptation may include designing a new scheduling policy based on analyzing backtest cases to achieve improved performance. As used herein, a “backtest case” refers to testing a machine learning (ML) model (e.g., a scheduling policy) that has been trained on historic data and evaluating the performance and/or effectiveness of the model in a deployment environment.

[0008] Aspects of some embodiments of the present disclosure provide for efficient policy adaptation, which may be implemented using limited computational resources. For example, a

production line having limited computational resources (e.g., using industrial computers) may be enabled to determine an improved scheduling policy based on limited processing capabilities and within a restricted time frame.

[0009] According to some embodiments of the present disclosure, a method for scheduling includes determining a second state distribution based on executing a first scheduling policy in a second environment, generating a second scheduling policy based on a first policy gradient associated with a first state distribution and the second state distribution, the first state distribution being associated with the first scheduling policy and a first environment, different from the second environment, and controlling a scheduling process based on the second scheduling policy.

[0010] The method may further include determining the first state distribution based on executing the first scheduling policy in the first environment.

[0011] The first environment may include a training environment in which the first scheduling policy is trained.

[0012] The first scheduling policy may be trained based on historic data.

[0013] The second environment may include a deployment environment, and the first scheduling policy may be tested in the deployment environment based on present data that differs from the historic data based on at least one parameter.

[0014] The method may further include determining a first performance associated with executing the first scheduling policy in the second environment, and determining to perform the controlling of the scheduling process based on the second scheduling policy based on comparing a second performance, associated with executing the second scheduling policy in the second environment, with the first performance.

[0015] The method may further include determining to perform the controlling of the scheduling process based on the second scheduling policy based on determining that a current time value is greater than or equal to a threshold time value.

[0016] The method may further include, based on executing the first scheduling policy in the second environment, determining a trajectory associated with the first scheduling policy.

[0017] The controlling the scheduling process based on the second scheduling policy may include one of changing an order of machine operations, selecting different navigation tasks, or changing an order of language model tasks.

[0018] The method may further include determining that a current time value is less than a threshold time value, determining a third state distribution associated with the second scheduling policy, and generating a third scheduling policy based on a second policy gradient associated with the second state distribution and the third state distribution.

[0019] According to some other embodiments of the present disclosure, a system for scheduling includes a processing circuit communicatively coupled to a machine, the processing circuit configured to perform determining a second state distribution based on executing a first scheduling policy in a second environment, generating a second scheduling policy based on a first policy gradient associated with a first state distribution and the second state distribution, the first state distribution being associated with the first scheduling policy and a first environment, different from the second environment, and controlling a scheduling process based on the second scheduling policy.

[0020] The processing circuit may be configured to perform determining the first state distribution based on executing the first scheduling policy in the first environment.

[0021] The first environment may include a training environment in which the first scheduling policy is trained.

[0022] The first scheduling policy may be trained based on historic data.

[0023] The second environment may include a deployment environment, and the first scheduling policy may be tested in the second environment based on updated data that differs from the historic data based on at least one parameter.

[0024] According to some other embodiments of the present disclosure, a system for scheduling includes a processing circuit and memory including instructions that, when executed by the processing circuit, cause the processing circuit to perform determining a second state distribution based on executing a first scheduling policy in a second environment, generating a second scheduling policy based on a first policy gradient associated with a first state distribution and the second state distribution, the first state distribution being associated with the first scheduling policy and a first environment, different from the second environment, and controlling a scheduling process based on the second scheduling policy.

[0025] The instructions, when executed by the processing circuit, may cause the processing circuit to perform determining the first state distribution based on executing the first scheduling policy in the first environment.

[0026] The first environment may include a training environment in which the first scheduling policy is trained.

[0027] The first scheduling policy may be trained based on historic data.

[0028] The second environment may include a deployment environment, and the first scheduling policy may be tested in the second environment based on updated data that differs from the historic data based on at least one parameter.

[0029] The above approaches improve on previous methods because they allow for efficient determination of improved scheduling policies that may be adapted to changing scenarios and evaluation metrics. For example, determining an improved scheduling policy, which previously took about 100 hours (e.g., based on trial-and-error alone) may take minutes or less based on aspects of some embodiments of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWING

[0030] In the following section, the aspects of the subject matter disclosed herein will be described with reference to exemplary embodiments illustrated in the figures.

[0031] FIG. 1A is a system diagram depicting a system for manufacturing scheduling, according to some embodiments of the present disclosure.

[0032] FIG. 1B is a block diagram depicting a method for determining a new scheduling policy, according to some embodiments of the present disclosure.

[0033] FIG. 2 is a block diagram depicting an RL decision process associated with determining new scheduling policies, according to some embodiments of the present disclosure.

[0034] FIG. 3 is a flowchart depicting example operations of a method for scheduling, according to some embodiments of the present disclosure.

[0035] FIG. 4 is a block diagram of an electronic device in a network environment, according to some embodiments of the present disclosure.

[0036] FIG. 5 is a flowchart of a method of executing a schedule (e.g., a scheduling policy), according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0037] In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the disclosure. It will be understood, however, by those skilled in the art that the disclosed aspects may be practiced without these specific details. In other instances, well-known methods, procedures, components and circuits have not been described in detail to not obscure the subject matter disclosed herein.

[0038] Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment may be included in at least one embodiment disclosed herein. Thus, the appearances of the phrases

“in one embodiment” or “in an embodiment” or “according to one embodiment” (or other phrases having similar import) in various places throughout this specification may not necessarily all be referring to the same embodiment. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner in one or more embodiments. In this regard, as used herein, the word “exemplary” means “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not to be construed as necessarily preferred or advantageous over other embodiments. Additionally, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. Also, depending on the context of discussion herein, a singular term may include the corresponding plural forms and a plural term may include the corresponding singular form. Similarly, a hyphenated term (e.g., “two-dimensional,” “pre-determined,” “pixel-specific,” etc.) may be occasionally interchangeably used with a corresponding non-hyphenated version (e.g., “two dimensional,” “predetermined,” “pixel specific,” etc.), and a capitalized entry (e.g., “Counter Clock,” “Row Select,” “PIXOUT,” etc.) may be interchangeably used with a corresponding non-capitalized version (e.g., “counter clock,” “row select,” “pixout,” etc.). Such occasional interchangeable uses shall not be considered inconsistent with each other.

[0039] Also, depending on the context of discussion herein, a singular term may include the corresponding plural forms and a plural term may include the corresponding singular form. It is further noted that various figures (including component diagrams) shown and discussed herein are for illustrative purpose only, and are not drawn to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity. Further, if considered appropriate, reference numerals have been repeated among the figures to indicate corresponding and/or analogous elements.

[0040] The terminology used herein is for the purpose of describing some example embodiments only and is not intended to be limiting of the claimed subject matter. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0041] It will be understood that when an element or layer is referred to as being on, “connected to” or “coupled to” another element or layer, it can be directly on, connected or coupled to the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly connected to” or “directly coupled to” another element or layer, there are no intervening elements or layers present. Like numerals refer to like elements throughout. As used herein, the terms “or” and “and/or” include any and all combinations of one or more of the associated listed items.

[0042] The terms “first,” “second,” etc., as used herein, are used as labels for nouns that they precede, and do not imply any type of ordering (e.g., spatial, temporal, logical, etc.) unless explicitly defined as such. Furthermore, the same reference numerals may be used across two or more figures to refer to parts, components, blocks, circuits, units, or modules having the same or similar functionality. Such usage is, however, for simplicity of illustration and ease of discussion only; it does not imply that the construction or architectural details of such components or units are the same across all embodiments or such commonly-referenced parts/modules are the only way to implement some of the example embodiments disclosed herein.

[0043] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this subject matter belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense

unless expressly so defined herein.

[0044] As used herein, the term “module” refers to any combination of software, firmware and/or hardware configured to provide the functionality described herein in connection with a module. For example, software may be embodied as a software package, code and/or instruction set or instructions, and the term “hardware,” as used in any implementation described herein, may include, for example, singly or in any combination, an assembly, hardwired circuitry, programmable circuitry, state machine circuitry, and/or firmware that stores instructions executed by programmable circuitry. The modules may, collectively or individually, be embodied as circuitry that forms part of a larger system, for example, but not limited to, an integrated circuit (IC), system on-a-chip (SoC), an assembly, and so forth.

[0045] FIG. 1A is a system diagram depicting a system for manufacturing scheduling, according to some embodiments of the present disclosure.

[0046] Referring to FIG. 1A, a system **1** (e.g., a manufacturing system) may include a processing circuit **120** (e.g., a processing circuit of the manufacturing system) for determining, updating, and/or implementing scheduling policies **130**. As used herein, a “scheduling policy” refers to any model that receives information about jobs and/or machines for processing the jobs and that outputs a job to be processed (e.g., outputs a rank of jobs to be processed). For example, a job having the highest rank may be processed by an available machine. An example of a scheduling policy is a machine learning (ML) model. For example, each scheduling policy **130** may include an ML model. The ML model may be any ML model known to one of ordinary skill in the art. In some embodiments, the ML model may include a neural network **131** (see FIG. 1B). In some embodiments, the ML model may include a linear model. As used herein “training” a scheduling policy refers to adjusting parameters (e.g., weights) of the ML model. The processing circuit **120** may correspond to the processor **420** of FIG. 4. The system **1** may include machines **M** (e.g., manufacturing devices, including robots and/or the like), products **P** (e.g., jobs), and resources **R**. In some embodiments, the processing circuit **120** may be communicatively coupled to the machines **M** to implement the scheduling policies **130** and/or to observe results (e.g., performance results) associated with the scheduling policies **130**. For example, the system **1** may include a first machine **M1** and a second machine **M2**. The machines **M** may utilize the resources **R** to produce the products **P**. The products may include, for example, different types of display devices or different types of integrated circuits (ICs). The products may include a first type of product **P1**, a second type of product **P2**, and a third type of product **P3**. The resources **R** may include, for example, a mask (e.g., a semiconductor photomask) and/or materials for producing the products **P**. Although FIG. 1A depicts only a few machines, products, and resources, it should be understood that many (e.g., thousands of) machines and processes may be used per product in a typical manufacturing system.

[0047] The performance of the system **1** may be determined based on one or more performance criteria (e.g., performance metrics). For example, performance metrics may include targets (e.g., goals) for production capacity (e.g., a number of products produced per day), efficiency (e.g., power consumption, minimized downtime, etc.), time (e.g., on-time delivery), and/or the like. As discussed above, a scheduling policy **130** may be trained in a training environment using a historic dataset. When tested in a deployment environment, the scheduling policy **130** may have different performance results from the training environment. For example, present data associated with present manufacturing scenarios may have changed from the historic data associated with previous manufacturing scenarios.

[0048] Aspects of some embodiments of the present disclosure allow the system **1** to improve (e.g., to optimize) scheduling policy performance in real time, with limited computational resources, within a short time frame, and without extensive tuning.

[0049] FIG. 1B is a block diagram depicting a method for determining a new scheduling policy, according to some embodiments of the present disclosure.

[0050] Referring to FIG. 1B, aspects of embodiments of the present disclosure allow for efficient adaptation of scheduling policies **130**. In some embodiments, one or more of the scheduling policies **130** may include an ML model. For example, the ML model may include a neural network **131**. Although the present disclosure depicts neural networks **131**, one or more of the scheduling policies **130** may include any suitable model known to one of ordinary skill in the art. For example, one or more of the scheduling policies **130** may include a linear model, a regression model (e.g., a neural network, a tree model, etc.), and/or the like. One or more processing circuits **120** may perform methods for generating a second scheduling policy **130k** (e.g., a k-th scheduling policy) based on updating a first scheduling policy **130a** (e.g., a previous scheduling policy). In some embodiments, the first scheduling policy **130a**, which may also be referred to mathematically as $\pi_{\text{sub.}\theta}$, may be trained in a first environment **100** (e.g., a “training environment”). The first environment **100** may train the first scheduling policy **130a** with a historic dataset HD including historic data. The historic dataset HD may include data corresponding to various parameters associated with scenarios (e.g., previously observed scenarios) of the system **1** (see FIG. 1A). In some embodiments, the first scheduling policy **130a** may be executed in the first environment **100**. In some embodiments, the first scheduling policy **130a** may be trained with a first processing circuit **120a**. The first processing circuit **120a** may have specialized processing resources. For example, the first processing circuit **120a** may include processing resources (e.g., one or more GPUs) that are suitable for artificial intelligence (AI) related processing. In some embodiments, a first Q-value **102** and/or a first state distribution **104** (e.g., a first steady-state distribution) may be obtained (e.g., determined) based on executing the first scheduling policy **130a** in the first environment **100**. For example, the Q-value **102** may be obtained through training an RL agent (see FIG. 2) in a training Markov decision process (MDP), while the first state distribution **104** may be obtained by counting a number of visitance (e.g., a number of visits) of a state through executing the first scheduling policy **130**. The first state distribution **104** may be represented mathematically as $d_{\text{sub.MDP.sub.old}}(s, a)$.

[0051] In some embodiments, the one or more processing circuits **120** may execute the first scheduling policy **130a** in a second environment **200** (e.g., a “deployment environment” or “testing environment”). The deployment environment **200** may test the first scheduling policy **130a** based on a present dataset PD including present data. The present dataset PD may include data corresponding to various parameters associated with scenarios (e.g., presently observed scenarios) of the system **1** (see FIG. 1A). Due to changing circumstances, the present dataset PD may differ from the historic dataset HD based on (e.g., with respect to) one or more parameters. For example, a parameter of the historic dataset HD may indicate that a first product P1 (see FIG. 1A) has an arrival frequency of **10** pieces per week, while the present dataset PD indicates that the first product P1 has an arrival frequency of 50 pieces per week. Accordingly, the first scheduling policy **130a** may perform differently in the first environment **100** from how the first scheduling policy **130a** performs in the second environment **200**. In some embodiments, the first scheduling policy **130a** may be tested with a second processing circuit **120b**. The second processing circuit **120b** may have more limited processing resources (e.g., more limited computational resources) than the specialized processing resources of the first processing circuit **120a**. For example, the second processing circuit **120b** may include general-purpose processing resources (e.g., one or more CPUs) that are less suitable for AI-related processing than the specialized processing resources of the first processing circuit **120a**. In some embodiments, a second Q-value **202** and/or a second state distribution **204** (e.g., a second steady-state distribution) may be obtained (e.g., determined) based on executing the first scheduling policy **130a** in the second environment **200**. The second state distribution **204** may be represented mathematically as $d_{\text{sub.MDP.sub.new}}(s, a)$. A first performance **206**, which may be represented mathematically as $G_{\text{sub.}\theta}$, and a first trajectory **208**, which may be represented mathematically as $\tau_{\text{sub.}\theta}$, may be obtained (e.g., determined) based on executing the first scheduling policy **130a** in the second environment **200**.

[0052] In some embodiments, the second scheduling policy **130k**, which may be represented mathematically as $\pi.\text{sub}.\theta'$, may be determined based on calculating a first policy gradient **210**, which may be associated with a state distribution difference, and which may be represented mathematically as $\nabla.\text{sub}.\theta J$. The first policy gradient **210** may be associated with the first state distribution **104** and the second state distribution **204**. In other words, the first policy gradient **210** may be calculated based a ratio of the second state distribution **204** and the first state distribution **104**. Determining the ratio of the second state distribution **204** and the first state distribution **104** may be referred to as “importance sampling” because the ratio of the second state distribution **204** and the first state distribution provides a way to determine (e.g., to calculate) an impact of the second state distribution **204** on a parameter (or parameters) being estimated. For example, the importance sampling involves changing a probability density function to generate more events (e.g., more rare events) that have more of an impact on the parameter (or parameters) being estimated. A state that is visited more frequently (e.g., a state that has a greater state distribution) may have a greater impact on the parameter (or parameters) being estimated.

[0053] In some embodiments, the second scheduling policy **130k** may be determined based on an update function **220**. The update function **220** may update the first scheduling policy **130a** with the first policy gradient **210**. For example, the second scheduling policy **130k** may correspond to the sum of the first scheduling policy **130a** and the first policy gradient **210**. The Q value **202** and the trajectory **208** may also be used to calculate the first policy gradient **210**, as discussed below in relation to, for example, equation 0.4 to equation 0.7.

[0054] In some embodiments, the second scheduling policy **130k** may be executed in the second environment **200**. A second performance **306**, which may be represented mathematically as $G.\text{sub}.\theta'$, may be determined based on executing the second scheduling policy **130k** in the second environment **200**. In some embodiments, the second scheduling policy **130k** may be selected to control a scheduling process, associated with the system **1** (see FIG. 1A), based on comparing the second performance **306** with the first performance **206**. For example, if the second performance **306** sufficiently improves the scheduling process (e.g., satisfies a given set of evaluation metrics), the second scheduling policy **130k** may be selected to control the scheduling process.

[0055] In some embodiments, the second scheduling policy **130k** or a third scheduling policy **130k+1** (e.g., a k+1-th scheduling policy) may be selected to control the scheduling process based on determining whether a current time value is greater than or equal to a threshold time value. For example, in some embodiments, the second processing circuit **120b** may determine a scheduling policy for controlling the scheduling process based on an iterative process **333**. For example, as part of the iterative process **333**, the second processing circuit **120b** may determine that the current time value is less than the threshold time value and may execute the second scheduling policy **130k** in the second environment **200** to determine a third state distribution **304**. The second processing circuit **120b** may generate the third scheduling policy **130k+1** based determining a second policy gradient associated with the second state distribution **204** and the third state distribution **304**. In other words, the scheduling policy may be updated iteratively based on determining whether more time remains to perform updating.

[0056] A process flow of the method for scheduling may include one or more of the following operations. A test policy may include the equation $1.0 * (\text{waiting time}) + 0.0001 * (\text{queue length})$. That is, the test policy μ may be trained to have a weight of 1.0 assigned to a waiting time (e.g., a waiting time for different products P in respective queues) and to have a weight of 0.0001 assigned to a queue length (e.g., a queue length of each of the products P in their respective queues). The processing circuit **120** may obtain a state distribution $d.\text{sub}.\text{MDP}.\text{sub}.\text{old}(s, a)$ of the test policy μ from its training environment. The processing circuit **120** may obtain a state distribution $d.\text{sub}.\text{MDP}.\text{sub}.\text{new}(s, a)$ and a corresponding schedule performance $G.\text{sub}.\text{1}$ of the test policy μ in a deployment environment. The processing circuit **120** may calculate a policy gradient of the test policy μ in the deployment environment. The processing circuit **120** may update the test policy μ ,

such that an updated test policy μ' includes the equation $0.8 * (\text{waiting time}) + 0.15 * (\text{queue length})$. That is, the weights respectively assigned to the waiting time and queue length may be changed based on the policy gradient. The processing circuit **120** may obtain the state distribution of $d.\text{sub.MDP.sub.new}, \mu'(s, a)$ with μ' and the corresponding schedule performance $G.\text{sub.2}$. The processing circuit **120** may output the better of the two schedules μ and μ' .

[0057] While embodiments of the present disclosure are described herein in the context of generating scheduling policies for coordinating various manufacturing processes and the like for the system **1**, the present disclosure is not limited thereto. For example, the systems and methods described herein may be applicable to any suitable systems and methods that may benefit from generating a new policy to achieve satisfactory results. In other words, as long as there is a policy that takes in some parameters/factors and outputs a decision/action, it can be used to generate a new policy according to one or more embodiments of the present disclosure. For example, the systems and methods described in more detail herein may be applicable to various robotic control applications, navigation systems, autonomous driving applications, large language model training, and the like.

[0058] FIG. **2** is a block diagram depicting an RL decision process associated with determining new scheduling policies, according to some embodiments of the present disclosure.

[0059] Referring to FIG. **2**, the processing circuit **120** may be configured to train scheduling policies **130** based on reinforcement learning. As used herein, “reinforcement learning” (RL) refers to a machine-learning technique, wherein an agent **40** (e.g., a machine-learning model) takes an action **50** (e.g., makes an update to one or more weights associated with the machine-learning model) in an environment **60** to improve or maximize a reward **70**. The action **50** may change a state **62** associated with the environment **60** to improve or increase the reward **70** achieved with respect to the state **62** (e.g., a previous state) of the system in association with the environment **60**. Based on RL, the processing circuit **120** may discover the actions **50** that maximize the reward **70** associated with any given state **62**.

[0060] RL for improving scheduling policies **130** may be modeled as a MDP denoted as $\langle S, A, R, P \rangle$, wherein an output scheduling policy **130** (or π) may be determined based on: a set of states **62** (e.g., environment-and-agent states S), a set of actions **50** (e.g., actions A of the agent **40**), a probability P of transitioning from a first state s to a second state s' under a given action a ; and an immediate reward $R.\text{sub.a}(s, s')$ after transitioning from the first state s to the second state s' under the given action a .

[0061] The following discussion provides mathematical notation corresponding to the disclosure above related to FIGS. **1B** and **2**.

[0062] An RL objective, which may be to achieve an optimized policy, may be represented mathematically as:

$$[00001] \max \quad \text{Math.}_{t=0}^T E_{S_t \sim d(s), a_t \sim (a.\text{Math. } s)} [\gamma^t r(S_t, a_t)] \quad (\text{equation } 0.1)$$

wherein $d.\text{sup.}\pi(s)$ refers to a steady-state distribution of states s , wherein $s.\text{sub.t}\tilde{d}.\text{sup.}\pi(s)$ refers to a steady-state distribution of the states, wherein $\pi(a|s)$ refers to an optimal scheduling policy, wherein $\gamma.\text{sup.t}$ refers to a discount factor, and wherein $[\gamma.\text{sup.tr}(s.\text{sub.t}, a.\text{sub.t})]$ refers to a cumulative discounted reward. The cumulative discounted reward $[\gamma.\text{sup.tr}(s.\text{sub.t}, a.\text{sub.t})]$ may be obtained given the steady-state distribution of the states $s.\text{sub.t}\tilde{d}.\text{sup.}\pi(s)$ and the optimal scheduling policy $\pi(a|s)$.

[0063] During training, the optimal scheduling policy $\pi(a|s)$ may be determined for the training environment based on achieving the steady-state distribution of $d.\text{sup.}\pi(s)$. However, during a deployment in a deployment environment an expectation associated with the determined optimal scheduling policy may no longer hold due to a distribution drift. The distribution drift may occur due to changing scenarios that differ from training data. For example, the deployment environment may include different types of orders (e.g., different types of product orders) than the types of

orders associated with the training data.

[0064] In some embodiments, a higher performance on the deployment case (e.g., in the deployment environment) with limited computational resources and time may be achieved by maximizing a summation of the cumulative discounted reward $[\gamma \cdot \text{sup.tr}(s_{\text{sub.t}}, a_{\text{sub.t}})]$ based on the deployment environment, which may be represented mathematically as:

$$[00002] \text{.Math. } {}^t r(s_t, a_t) \sim \text{.Math. } S', A, R, P', s_0 \text{.Math. (equation0.2)}$$

[0065] That is, the deployment environment $\langle S', A, R, P' \rangle$ may differ from the training environment based on S' and P' . In some embodiments, the training environment (e.g., the old MDP environment) may have a continuous arrival of jobs, while the jobs for the deployment environment (e.g., the new MDP environment) may be known in advance.

[0066] A policy gradient method may be utilized to determine an optimized policy in the deployment environment. In some embodiments, a policy gradient target may be determined based on:

$$[00003] J = E \sim [\text{.Math. } {}^T_{t=0} {}^t r(s_t, a_t)] \text{ and (equation0.3)}$$

$$\nabla J = \frac{1}{N} \cdot \text{.Math. } {}^N_{i=0} \cdot \text{.Math. } {}^T_{t=0} \nabla {}^t \log (a_{t,i} \cdot \text{.Math. } s_{t,i}) \hat{Q}(s_{t,i}, a_{t,i}) \text{ (equation0.4)}$$

wherein $J_{\text{sub.}\theta}$ refers to a reward (e.g., a final reward), wherein

$E_{\text{sub.}\tau} \sim \pi_{\text{sub.}\theta}[\Sigma_{\text{sub.t}=0}^{\text{sup.T}} \gamma \cdot \text{sup.tr}(s_{\text{sub.t}}, a_{\text{sub.t}})]$ refers to an expected summation of rewards (e.g., immediate rewards), $\nabla_{\text{sub.}\theta} J$ refers to the policy gradient, N refers to a number of rollouts (e.g., a number of training times), $\log \pi_{\text{sub.}\theta}(a_{\text{sub.t},i} | s_{\text{sub.t},i})$ refers to an expectation value of a log of the policy, and $\{\text{circumflex over (Q)}\}(s_{\text{sub.t},i}, a_{\text{sub.t},i})$ refers to a Q value of the state and action pair.

[0067] In deployment environments having short time limits, the environment may be considered static (e.g., not changing) and the number of rollouts N may, thus, be equal to one. Accordingly, the gradient may be determined mathematically based on:

$$[00004] \nabla J = \text{.Math. } {}^T_{t=0} \nabla {}^t \log (a_t \cdot \text{.Math. } s_t) \hat{Q}(s_t, a_t) \text{ (equation0.5)}$$

[0068] In some embodiments, a modified version of equation 0.4 above for determining the policy gradient target may be represented mathematically as (equation 0.6):

$$[00005] \nabla J = \frac{1}{N} \cdot \text{.Math. } \frac{(\text{ }_{i \sim \text{MDP}_{\text{new}}})}{(\text{ }_{i \sim \text{MDP}_{\text{old}}})} \cdot \text{.Math. } {}^T_{t=0} \nabla {}^t \log (a_{t,i} \cdot \text{.Math. } s_{t,i}) \hat{Q}(s_{t,i}, a_{t,i})$$

[0069] In some embodiments, a computational cost (e.g., an exponential computational cost) associated with determining:

$$[00006] (\text{ }_{i \sim \text{MDP}_{\text{new}}}) (\text{ }_{i \sim \text{MDP}_{\text{old}}}) \text{ (equation0.7)}$$

may be reduced (e.g., minimized) by estimating the ratio of the state distribution of the new MDP environment (e.g., the deployment environment) and the state distribution of the old MDP environment (e.g., the training environment):

$$[00007] \frac{d_{\text{MDP}_{\text{new}}}(s, a)}{d_{\text{MDP}_{\text{old}}}(s, a)} \text{ (equation0.8)}$$

given the consistency condition. For example, given that the deployment environment is static, a computation cost associated with determining the policy gradient target may be reduced, such that the policy gradient target may be determined based on:

$$[00008] \nabla J = \text{.Math. } {}^T_{t=0} \nabla {}^t \log (a_{t,i} \cdot \text{.Math. } s_{t,i}) \frac{d_{\text{MDP}_{\text{new}}}(s, a)}{d_{\text{MDP}_{\text{old}}}(s, a)} \hat{Q}(s_{t,i}, a_{t,i}) \text{ (equation1.)}$$

[0070] FIG. 3 is a flowchart depicting operations of a method for scheduling, according to some embodiments of the present disclosure.

[0071] Referring to FIG. 3, a method 300 for scheduling, as similarly discussed above with reference to FIG. 1B, may include one or more of the following operations. A processing circuit 120 (see FIGS. 1A-2) may select a first scheduling policy for deployment (operation 301). In some embodiments, the first scheduling policy may be trained in a training environment based on historic

data. The processing circuit **120** may obtain the distribution (e.g., the steady-state distribution) associated with the training environment $d.sub.MDP.sub.old(s, a)$ (operation **302**). The processing circuit **120** may rollout the trajectory using the first scheduling policy (operation **303**). For example, the processing circuit **120** may execute the first scheduling policy in a deployment environment. The processing circuit **120** may identify the distribution of the first scheduling policy in the deployment environment $d.sub.MDP.sub.new(s, a)$ (operation **304**). The processing circuit **120** may update the scheduling policy based on the policy gradient $\square.sub.\theta J$, which, as discussed above, may be determined based on a ratio of the distribution of the first scheduling policy $d.sub.MDP.sub.new(s, a)$ in the deployment environment to the distribution of the first scheduling policy in the training environment $d.sub.MDP.sub.old(s, a)$ (operation **305**). The processing circuit **120** may rollout the trajectory with the updated policy based on the policy gradient $\nabla.sub.\theta J$ (operation **306**). For example, the processing circuit **120** may execute an updated scheduling policy in the deployment environment. The processing circuit **120** may determine whether a current time value T (e.g., an updated time value) is greater than or equal to a threshold time value $T.sub.thres$ (operation **307**). If the current time value is greater than or equal to the threshold time value, the processing circuit **120** may obtain the best performing scheduling policy based on comparing performances of the scheduling policies (operation **308**). If the current time value is less than the threshold time value, the processing circuit **120** may perform an iterative process of determining state distributions of new policies and determining corresponding policy gradients as in operations **304** through **306** until the current time value is greater than or equal to the threshold time value. Once the new scheduling policy (e.g., the best performing scheduling policy) is obtained, the processing circuit **120** may use the new scheduling policy to control a scheduling process of the system **1** (see FIG. **1**).

[0072] In some embodiments, the new scheduling policy may be executed to coordinate machine actions in a factory, schedule navigation tasks, coordinate language model tasks, and/or the like. For example, executing the new scheduling policy, and controlling a scheduling process, may include changing the order of machine operations, selecting different navigation tasks, changing an order of language model tasks, and/or the like, based on the new scheduling-policy schedule.

[0073] The method **300** may be performed on one or more components of the electronic device **401** of FIG. **4**. For example, one or more operations may be performed using a combination of software components and hardware components corresponding to the processor **420** and the memory **430** of FIG. **4**.

[0074] FIG. **4** is a block diagram of an electronic device in a network environment **400**, according to some embodiments of the present disclosure.

[0075] Referring to FIG. **4**, an electronic device **401** in a network environment **400** may communicate with an electronic device **402** via a first network **498** (e.g., a short-range wireless communication network), or an electronic device **404** or a server **408** via a second network **499** (e.g., a long-range wireless communication network). The electronic device **401** may communicate with the electronic device **404** via the server **408**. The electronic device **401** may include a processor **420**, a memory **430**, an input device **450**, a sound output device **455**, a display device **460**, an audio module **470**, a sensor module **476**, an interface **477**, a haptic module **479**, a camera module **480**, a power management module **488**, a battery **489**, a communication module **490**, a subscriber identification module (SIM) card **496**, or an antenna module **497**. In one embodiment, at least one (e.g., the display device **460** or the camera module **480**) of the components may be omitted from the electronic device **401**, or one or more other components may be added to the electronic device **401**. Some of the components may be implemented as a single integrated circuit (IC). For example, the sensor module **476** (e.g., a fingerprint sensor, an iris sensor, or an illuminance sensor) may be embedded in the display device **460** (e.g., a display).

[0076] The processor **420** may execute software (e.g., a program **440**) to control at least one other component (e.g., a hardware or a software component) of the electronic device **401** coupled with

the processor **420** and may perform various data processing or computations.

[0077] As at least part of the data processing or computations, the processor **420** may load a command or data received from another component (e.g., the sensor module **476** or the communication module **490**) in volatile memory **432**, process the command or the data stored in the volatile memory **432**, and store resulting data in non-volatile memory **434**. The processor **420** may include a main processor **421** (e.g., a central processing unit (CPU) or an application processor (AP)), and an auxiliary processor **423** (e.g., a graphics processing unit (GPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **421**. Additionally or alternatively, the auxiliary processor **423** may be adapted to consume less power than the main processor **421**, or execute a particular function. The auxiliary processor **423** may be implemented as being separate from, or a part of, the main processor **421**.

[0078] The auxiliary processor **423** may control at least some of the functions or states related to at least one component (e.g., the display device **460**, the sensor module **476**, or the communication module **490**) among the components of the electronic device **401**, instead of the main processor **421** while the main processor **421** is in an inactive (e.g., sleep) state, or together with the main processor **421** while the main processor **421** is in an active state (e.g., executing an application). The auxiliary processor **423** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **480** or the communication module **490**) functionally related to the auxiliary processor **423**.

[0079] The memory **430** may store various data used by at least one component (e.g., the processor **420** or the sensor module **476**) of the electronic device **401**. The various data may include, for example, software (e.g., the program **440**) and input data or output data for a command related thereto. The memory **430** may include the volatile memory **432** or the non-volatile memory **434**. Non-volatile memory **434** may include internal memory **436** and/or external memory **438**.

[0080] The program **440** may be stored in the memory **430** as software, and may include, for example, an operating system (OS) **442**, middleware **444**, or an application **446**.

[0081] The input device **450** may receive a command or data to be used by another component (e.g., the processor **420**) of the electronic device **401**, from the outside (e.g., a user) of the electronic device **401**. The input device **450** may include, for example, a microphone, a mouse, or a keyboard.

[0082] The sound output device **455** may output sound signals to the outside of the electronic device **401**. The sound output device **455** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or recording, and the receiver may be used for receiving an incoming call. The receiver may be implemented as being separate from, or a part of, the speaker.

[0083] The display device **460** may visually provide information to the outside (e.g., a user) of the electronic device **401**. The display device **460** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. The display device **460** may include touch circuitry adapted to detect a touch, or sensor circuitry (e.g., a pressure sensor) adapted to measure the intensity of force incurred by the touch.

[0084] The audio module **470** may convert a sound into an electrical signal and vice versa. The audio module **470** may obtain the sound via the input device **450** or output the sound via the sound output device **455** or a headphone of an external electronic device **402** directly (e.g., wired) or wirelessly coupled with the electronic device **401**.

[0085] The sensor module **476** may detect an operational state (e.g., power or temperature) of the electronic device **401** or an environmental state (e.g., a state of a user) external to the electronic device **401**, and then generate an electrical signal or data value corresponding to the detected state. The sensor module **476** may include, for example, a gesture sensor, a gyro sensor, an atmospheric

pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0086] The interface **477** may support one or more specified protocols to be used for the electronic device **401** to be coupled with the external electronic device **402** directly (e.g., wired) or wirelessly. The interface **477** may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0087] A connecting terminal **478** may include a connector via which the electronic device **401** may be physically connected with the external electronic device **402**. The connecting terminal **478** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

[0088] The haptic module **479** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or an electrical stimulus which may be recognized by a user via tactile sensation or kinesthetic sensation. The haptic module **479** may include, for example, a motor, a piezoelectric element, or an electrical stimulator.

[0089] The camera module **480** may capture a still image or moving images. The camera module **480** may include one or more lenses, image sensors, image signal processors, or flashes. The power management module **488** may manage power supplied to the electronic device **401**. The power management module **488** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0090] The battery **489** may supply power to at least one component of the electronic device **401**. The battery **489** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0091] The communication module **490** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **401** and the external electronic device (e.g., the electronic device **402**, the electronic device **404**, or the server **408**) and performing communication via the established communication channel. The communication module **490** may include one or more communication processors that are operable independently from the processor **420** (e.g., the AP) and supports a direct (e.g., wired) communication or a wireless communication. The communication module **490** may include a wireless communication module **492** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **494** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **498** (e.g., a short-range communication network, such as BLUETOOTH™, wireless-fidelity (Wi-Fi) direct, or a standard of the Infrared Data Association (IrDA)) or the second network **499** (e.g., a long-range communication network, such as a cellular network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single IC), or may be implemented as multiple components (e.g., multiple ICs) that are separate from each other. The wireless communication module **492** may identify and authenticate the electronic device **401** in a communication network, such as the first network **498** or the second network **499**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **496**.

[0092] The antenna module **497** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **401**. The antenna module **497** may include one or more antennas, and, therefrom, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **498** or the second network **499**, may be selected, for example, by the communication module **490** (e.g., the wireless communication module **492**). The signal or the power may then be transmitted or received between

the communication module **490** and the external electronic device via the selected at least one antenna.

[0093] Commands or data may be transmitted or received between the electronic device **401** and the external electronic device **404** via the server **408** coupled with the second network **499**. Each of the electronic devices **402** and **404** may be a device of a same type as, or a different type, from the electronic device **401**. All or some of operations to be executed at the electronic device **401** may be executed at one or more of the external electronic devices **402**, **404**, or the server **408**. For example, if the electronic device **401** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **401**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request and transfer an outcome of the performing to the electronic device **401**. The electronic device **401** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, or client-server computing technology may be used, for example.

[0094] FIG. **5** is a flowchart of a method of executing a schedule (e.g., a scheduling policy), according to some embodiments of the present disclosure.

[0095] The method **500** shown in FIG. **5**, may be performed, for example, by the processing circuit **120** described above with reference to FIGS. **1A-2**. However, the present disclosure is not limited thereto, and the operations shown in the method **500** may be performed by any suitable one of the components and elements or any suitable combination of the components and elements of those of one or more embodiments described above. Further, the present disclosure is not limited to the sequence or number of the operations of the method **500** shown in FIG. **5**, and can be altered into any desired sequence or number of operations as recognized by a person having ordinary skill in the art. For example, in some embodiments, the order may vary, or the method **500** may include fewer or additional operations. Further, the operations shown in method **500** may be performed sequentially, or at least some of the operations thereof may be performed concurrently (e.g., simultaneously, or substantially simultaneously).

[0096] Referring to FIG. **5**, the method **500** may start, and a schedule (e.g., a scheduling policy) may be executed (operation **505**). For example, in some embodiments, the schedule may be executed to coordinate machine actions in a simulation of a factory, schedule navigation tasks, coordinate language model tasks, and the like.

[0097] A policy state distribution of an environment used to train the scheduling policy associated with the schedule may be obtained (e.g., received) (operation **510**). A policy state distribution of the environment associated with the simulation of the factory, navigation tasks, language model, and/or the like may be obtained (e.g., received) (operation **515**). The schedule may be updated per the ratio of the state distributions (operation **520**).

[0098] The new schedule may be executed (operation **525**), and the method **500** may end. For example, executing the new schedule may include changing the order of machine operations, selecting different navigation tasks, changing an order of language model tasks, and the like, based on the new schedule.

[0099] Embodiments of the subject matter and the operations described in this specification may be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Embodiments of the subject matter described in this specification may be implemented as one or more computer programs, i.e., one or more modules of computer-program instructions, encoded on computer-storage medium for execution by, or to control the operation of data-processing apparatus. Alternatively or additionally, the program

instructions can be encoded on an artificially-generated propagated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal, which is generated to encode information for transmission to suitable receiver apparatus for execution by a data processing apparatus. A computer-storage medium can be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial-access memory array or device, or a combination thereof. Moreover, while a computer-storage medium is not a propagated signal, a computer-storage medium may be a source or destination of computer-program instructions encoded in an artificially-generated propagated signal. The computer-storage medium can also be, or be included in, one or more separate physical components or media (e.g., multiple CDs, disks, or other storage devices). Additionally, the operations described in this specification may be implemented as operations performed by a data-processing apparatus on data stored on one or more computer-readable storage devices or received from other sources.

[0100] While this specification may contain many specific implementation details, the implementation details should not be construed as limitations on the scope of any claimed subject matter, but rather be construed as descriptions of features specific to particular embodiments. Certain features that are described in this specification in the context of separate embodiments may also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment may also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

[0101] Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

[0102] Thus, particular embodiments of the subject matter have been described herein. Other embodiments are within the scope of the following claims. In some cases, the actions set forth in the claims may be performed in a different order and still achieve desirable results. Additionally, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing may be advantageous.

[0103] As will be recognized by those skilled in the art, the innovative concepts described herein may be modified and varied over a wide range of applications. Accordingly, the scope of claimed subject matter should not be limited to any of the specific exemplary teachings discussed above, but is instead defined by the following claims.

Claims

1. A method for scheduling, the method comprising: determining a second state distribution based on executing a first scheduling policy in a second environment, generating a second scheduling policy based on a first policy gradient associated with a first state distribution and the second state distribution, the first state distribution being associated with the first scheduling policy and a first environment, different from the second environment; and controlling a scheduling process based on the second scheduling policy.

2. The method of claim 1, further comprising determining the first state distribution based on

executing the first scheduling policy in the first environment.

3. The method of claim 1, wherein the first environment comprises a training environment in which the first scheduling policy is trained.

4. The method of claim 3, wherein the first scheduling policy is trained based on historic data.

5. The method of claim 4, wherein: the second environment comprises a deployment environment; and the first scheduling policy is tested in the deployment environment based on present data that differs from the historic data based on at least one parameter.

6. The method of claim 1, further comprising: determining a first performance associated with executing the first scheduling policy in the second environment; and determining to perform the controlling of the scheduling process based on the second scheduling policy based on comparing a second performance, associated with executing the second scheduling policy in the second environment, with the first performance.

7. The method of claim 1, further comprising determining to perform the controlling of the scheduling process based on the second scheduling policy based on determining that a current time value is greater than or equal to a threshold time value.

8. The method of claim 1, further comprising, based on executing the first scheduling policy in the second environment, determining a trajectory associated with the first scheduling policy.

9. The method of claim 1, wherein the controlling the scheduling process based on the second scheduling policy comprises one of changing an order of machine operations, selecting different navigation tasks, or changing an order of language model tasks.

10. The method of claim 1, further comprising: determining that a current time value is less than a threshold time value; determining a third state distribution associated with the second scheduling policy; and generating a third scheduling policy based on a second policy gradient associated with the second state distribution and the third state distribution.

11. A system for scheduling, the system comprising: a processing circuit communicatively coupled to a machine, the processing circuit configured to perform: determining a second state distribution based on executing a first scheduling policy in a second environment, generating a second scheduling policy based on a first policy gradient associated with a first state distribution and the second state distribution, the first state distribution being associated with the first scheduling policy and a first environment, different from the second environment; and controlling a scheduling process based on the second scheduling policy.

12. The system of claim 11, wherein the processing circuit is configured to perform determining the first state distribution based on executing the first scheduling policy in the first environment.

13. The system of claim 11, wherein the first environment comprises a training environment in which the first scheduling policy is trained.

14. The system of claim 13, wherein the first scheduling policy is trained based on historic data.

15. The system of claim 14, wherein: the second environment comprises a deployment environment; and the first scheduling policy is tested in the second environment based on updated data that differs from the historic data based on at least one parameter.

16. A system for scheduling, the system comprising: a processing circuit and memory comprising instructions that, when executed by the processing circuit, cause the processing circuit to perform: determining a second state distribution based on executing a first scheduling policy in a second environment, generating a second scheduling policy based on a first policy gradient associated with a first state distribution and the second state distribution, the first state distribution being associated with the first scheduling policy and a first environment, different from the second environment; and controlling a scheduling process based on the second scheduling policy.

17. The system of claim 16, wherein the instructions, when executed by the processing circuit, cause the processing circuit to perform determining the first state distribution based on executing the first scheduling policy in the first environment.

18. The system of claim 16, wherein the first environment comprises a training environment in

which the first scheduling policy is trained.

19. The system of claim 18, wherein the first scheduling policy is trained based on historic data.

20. The system of claim 19, wherein: the second environment comprises a deployment environment; and the first scheduling policy is tested in the second environment based on updated data that differs from the historic data based on at least one parameter.
